

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	MTO for Long Lead Items (Piping/Mechanical/Pipeline)
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MTO for Long Lead Items (Piping/Mech./Pipeline)

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Revision History

Rev No.	Date	Description of Revision
R01	01.08.22	Issued for Review
R02		
A01		

MTO for Long Lead Items (Piping/Mechanical/Pipeline)**Revision Philosophy:**

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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LONG LEAD ITEMS FOR PIPING/MECHANICAL-BBOU MANIFOLD STATION													
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE
PIPES (BBOU MANIFOLD STATION)													
1	PIPE	C70	24"	--	24.2mm	--	900#	ASME B36.10/B36.19,	SEAMLESS/WELDED, 100% RADIOGRAPHY, ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 11 (12m SRL)	m	16.9	7.1	24
2	PIPE	C70	2"	--	80S	--	900#	ASME B36.10/B36.19,	SEAMLESS/WELDED, 100% RADIOGRAPHY, ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 1 (6m SRL)	m	13.6	4.4	18
3	PIPE	C03	24"	--	100	--	900#	ASME B36.10M	WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	70.5	1.5	72
4	PIPE	C03	8"	--	120	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	30.2	5.8	36
5	PIPE	C03	6"	--	120	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	29.3	0.7	30
6	PIPE	C03	4"	--	120	--	900#	ASME B36.10M	'SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	54.9	5.1	60
7	PIPE	C03	2"	Items (Piping/M	160	--	900#	ASME B36.10M	'SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	36.4	5.6	42
8	PIPE	C03	3/4"	--	160	--	900#	ASME B36.10M	'SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (6m SRL)	m	0.2	5.8	6
9	PIPE	B03	6"	--	STD	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	22.7	1.3	24
10	PIPE	B03	4"	--	STD	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	25.4	4.6	30
11	PIPE	B03	2"	--	XS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	1.3	4.7	6
12	PIPE	B03	3/4"	--	XS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	0.2	5.8	6
PIPING VALVES (BBOU MANIFOLD STATION)													
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	DESIG CODE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO PROCURE
1	24" SDV 1001	C03	BALL	--	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
1	24"	C03	BALL	--	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4
1	24"	C03	BALL	--	FB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	2	0	2
2	8"	C03	BALL	--	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4
3	6"	C03	BALL	--	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4
4	6"	B03	BALL	--	RB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB,TRUNNION MTD BALL, LEVER OPR .	PCS	6	0	6
5	4" BDV 1001	C03	BALL	--	FB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
5	4"	C03	BALL	--	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	13	0	13
6	4"	B03	BALL	--	RB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB,TRUNNION MTD BALL, LEVER OPR .	PCS	1	0	1
6	2"	D70	BALL	--	RB	RTJ FLANGED WITH SMOOTH FINISH	1500#		FLGD, RJ, CL1500, API SPEC 6D, ASTM A182 F51 OR ASTM A995 GR 4A BODY, UNS S31803 TRIM, SPB/TPE, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
7	2"	C03	BALL	--	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	21	0	21
8	2"	B03	BALL	--	RB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB,TRUNNION MTD BALL, LEVER OPR .	PCS	4	0	4
9	3/4"	C03	BALL	--		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	1500#		SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	10	0	10
10	3/4"	B03	BALL	--		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	800#		SW, CL800, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	4	0	4
11	4"	C03	GLOBE	--		RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	4	0	4
12	2"	C03	GLOBE	--		RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC.HNDWHL OPR W/ GEARBOX	PCS	7	0	7

LONG LEAD ITEMS FOR PIPELINE -BBOU													
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE
PIPES													
1	PIPE (Mainline from BBOU and ELEPA)	ASME B31.8	24"	--	15.88mm	--	900#	ASME B36.10M	API %L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 23.83mm (12m SRL)	m	6,700.0	1000.0	7,700.0
4	PIPE	ASME B31.8	24"	--	15.88mm	--	900#	ASME B36.10M	API %L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 23.83mm (12m SRL)	m	22.8	61.2	84
2	PIPE	ASME B31.8	2"	--	80/(5.54mm)	--	900#	API 5L/ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE X70 PSL 2, WT 5.54mm (12m SRL)	m	7.2	4.8	12
3	PIPE	ASME B31.8	3/4"	--	160	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	1.0	5.0	6
SPECIAL PIPING ITEMS													
18	INSULATING JOINT	ASME B31.8	24"	--	--	--	900#	ASME 31.3 ASME VIII DIV. 1	BODY: API 5L GR.B; PUPS: API 5L GR.B, PSL2, SAWL; FILLER ISOLATOR: EPOXY RESIN COLD CURED, COATING: AMINE-CURED EPOXY	PCS	2	0	2
PIPELINE VALVES -BBOU MANIFOLD													
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	BORE TYPE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO PROCURE
1	24" SDV 1002	ASME B31.8	BALL	--	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
1	24"	ASME B31.8	BALL	--	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	5	0	5
2	2"	ASME B31.8	BALL	--	RB	RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4
3	3/4"	ASME B31.8	BALL	--		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	1500#	--	SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	1	0	1
4	2"	ASME B31.8	GLOBE	--		RTJ FLANGED WITH SMOOTH FINISH	900#	--	FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC.HNDWHL OPR W/ GEARBOX	PCS	2	0	2

LONG LEAD ITEMS FOR PIPING/MECHANICAL- ELEPA MANIFOLD STATION													
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE
PIPES (ELEPA MANIFOLD STATION)													
1	PIPE	D70	36"	--	59.2mm	--	1500#	ASME B36.10/B36.19,	SEAMLESS/WELDED, 100% RADIOGRAPHY, ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 1 (12m SRL)	m	13.5	10.5	24
2	PIPE	D06	36"	--	76.1mm	--	1500#	ASME B36.10/B36.19,	WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE X60 PSL 2 (12m SRL)	m	4.0	8.0	12
3	PIPE	D06	2"	--	11.07mm	--	1500#	ASME B36.10/B36.19,	SEAMLESS/WELDED, 100% RADIOGRAPHY, ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 1 (12m SRL)	m	13.6	4.4	18
4	PIPE	C06	36"	--	38.2mm	--	900#	ASME B36.10M	WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE X60 PSL 2 (12m SRL)	m	62.1	9.9	72
5	PIPE	C06	24"	--	100	--	900#	ASME B36.10M	WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE X60 PSL 2 (12m SRL)	m	18.9	5.1	24
6	PIPE	C06	12"	--	140	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	31.4	4.6	36
7	PIPE	C06	8"	ms (Pipin	140	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	29.1	6.9	36
8	PIPE	C06	6"	--	160	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	135.2	2.8	138
9	PIPE	C06	4"	--	XXS	--	900#	ASME B36.10M	'SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	71.6	0.4	72
10	PIPE	C06	2"	--	XXS	--	900#	ASME B36.10M	'SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	28.6	1.4	30
11	PIPE	C06	3/4"	--	XXS	--	900#	ASME B36.10M	'SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (6m SRL)	m	1.0	5.0	6
12	PIPE	B06	6"	--	XS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	39.2	2.8	42
13	PIPE	B06	4"	--	120	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	47.0	1.0	48
14	PIPE	B06	2"	--	XXS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (12m SRL)	m	1.3	4.7	6
15	PIPE	B06	3/4"	--	XXS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	1.0	5.0	6

PIPING VALVES (ELEPA MANIFOLD)													
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	DESIGN CODE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO PROCURE
109	36" SDV 1001	D06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
110	36"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4
111	36"	C06	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	2	0	2
112	24"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	2	0	2
113	12"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4
114	8"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	2	0	2
115	6"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	7	0	7
116	6"	B06	BALL	—	FB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	7	0	7
117	4" BDV 1001	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
118	4"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	18	0	18
119	2"	D70	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	1500#		FLGD, RJ, CL1500, API SPEC 6D, ASTM A182 F51 OR ASTM A995 GR 4A BODY, UNS S31803 TRIM, SPB/TPE, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
120	2"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	24	0	24
121	2"	B06	BALL	—	RB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, FLOATINF BALL, LEVER OPR .	PCS	3	0	3
122	3/4"	C06	BALL	—		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	1500#		SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	10	0	10
123	3/4"	B06	BALL	—		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	800#		SW, CL800, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	5	0	5
124	4"	C06	GLOBE	—		RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	13	0	13
125	2"	C06	GLOBE	—		RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	9	0	9
126	2"	C06	BALL	—		RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	12	0	12

LONG LEAD ITEMS FOR PIPELINE - ELEPA													
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE
PIPES													
1	PIPE (Mainline from Elapa to AKabel)	ASME B31.8	36"	--	23.83mm	--	900#	API 5L / ASME B36.10M	API %L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 23.83mm (12m SRL)	m	31,500.0	1500.0	33,000
2	PIPE	ASME B31.8	36"	--	23.83mm	--	900#	API 5L / ASME B36.10M	API %L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 23.83mm (12m SRL)	m	74.2	9.8	84
3	PIPE	ASME B31.8	2"	--	80(5.54mm)	--	900#	API 5L / ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE X70 PSL 2, WT 5.54mm (6m SRL)	m	13.3	10.7	24
4	PIPE	ASME B31.8	24"	--	15.88mm	--	900#	ASME B36.10M	API %L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 15.88mm (12m SRL)	m	48.7	11.3	60
5	PIPE	ASME B31.8	3/4"	--	XXS	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	1.0	5.0	6
SPECIAL PIPING ITEMS													
27	INSULATING JOINT	ASME B31.8	36"	--	--	--	900#	ASME 31.8 ASME VIII DIV. 1	BODY: API 5L GR.B; PUPS: API 5L GR.B, PSL2, SAWL; FILLER ISOLATOR: EPOXY RESIN COLD CURED; COATING: AMINE-CURED EPOXY	PCS	2	0	2
28	INSULATING JOINT	ASME B31.8	24"	--	--	--	900#	ASME 31.3 ASME VIII DIV. 1	BODY: API 5L GR.B; PUPS: API 5L GR.B, PSL2, SAWL; FILLER ISOLATOR: EPOXY RESIN COLD CURED; COATING: AMINE-CURED EPOXY	PCS	1	0	1
PIPELINE VALVES - ELEPA													
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	DESIGN CODE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO PROCURE
29	36" SDV1002	ASME B31.8	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
30	36"	ASME B31.8	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4
31	24" SDV 1003	ASME B31.8	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A182 F51 OR ASTM A995 GR 4A BODY, UNS S31803 TRIM, SPB/TPE, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
32	24"	ASME B31.8	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A182 F51 OR ASTM A995 GR 4A BODY, UNS S31803 TRIM, SPB/TPE, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
33	2"	ASME B31.8	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR	PCS	4	0	4
34	3/4"	ASME B31.8	BALL	—		SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	1500#	—	SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	3	0	3
35	2"	ASME B31.8	GLOBE	—		RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	3	0	3

LONG LEAD ITEMS FOR PIPING/MECHANICAL- AKABEL PIG RECEIVER STATION													
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE
PIPES -AKABEL													
3	PIPE	C06	36"	—	38.2mm	--	900#	ASME B36.10M	WELDED, SAW 100% RADIOGRAPHY, ASME B36.10, API SPEC 5L GRADE X60 PSL 2 (12m SRL)	m	16.7	7.3	24
5	PIPE	C06	12"	—	140	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (12m SRL)	m	15.8	8.2	24
8	PIPE	C06	4"	—	XXS	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (6m SRL)	m	37.4	34.6	72
9	PIPE	C06	2"	—	XXS	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (6m SRL)	m	9.4	20.6	30
10	PIPE	C06	3/4"	—	XXS	--	900#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2 (6m SRL)	m	0.3	5.7	6
11	PIPE	B06	6"	—	XS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	28.8	7.2	36
12	PIPE	B06	4"	ms (Pipin	120	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	6.2	41.8	48
14	PIPE	B06	3/4"	—	XXS	--	150#	ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	0.1	5.9	6
PIPING VALVES -AKABEL													
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	DESIGN CODE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO PROCURE
1	36"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	1	0	1
3	12"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	2	0	2
6	6"	B06	BALL	—	FB	RF FLANGED WITH SMOOTH FINISH	150#		FLGD, RF, CL150, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	2	0	2
7	4"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	5	0	5
9	2"	C06	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	7	0	7
11	3/4"	C06	BALL	—	—	SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	1500#		SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	2	0	2
12	3/4"	B06	BALL	—	—	SW WITH 100mm PUP PIECE + 1/2" NPT-F WITH SS WELD OVERLAY	800#	—	SW, CL800, API SPEC 6D, A105N BODY, SS316 TRIM, FLTING BALL, LEVER OPR	PCS	2	0	2
13	4"	C06	GLOBE	—	—	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	2	0	2
14	2"	C06	GLOBE	—	—	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	3	0	3
15	2"	C06	DOUBLE BLOCK & BLEED	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#		FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR .	PCS	6	0	6

LONG LEAD ITEMS FOR PIPELINE - AKABEL													
S/NO	ITEM DESCRIPTION	MATERIAL CLASS	SIZE 1	SIZE 2	SCHEDULE / THICKNESS 1	SCHEDULE / THICKNESS 2	PRESSURE RATING	DESIGN CODE	MATERIAL DESCRIPTION	UNIT	QUANTITY REQUIRED	CONTINGENCY	QUANTITY TO PROCURE
PIPES													
1	PIPE	ASME B31.8	36"	--	23.83mm	--	900#	API 5L / ASME B36.10M	API 5L GR X70 PSL 2, SAWL, 100% RADIOGRAPHY, ASME B36.10, WT 23.83mm (12m SRL)	m	33.8	2.2	36
2	PIPE	ASME B31.8	2"	--	80/(5.54mm)	--	900#	API 5L / ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE X70 PSL 2, WT 5.54mm (6m SRL)	m	4.7	19.3	24
14	PIPE	ASME B31.8	3/4"	--	XXS	--	900#	API 5L / ASME B36.10M	SEAMLESS, BE, ASME B36.10, API SPEC 5L GRADE B PSL 2M (6m SRL)	m	1.0	5.0	6
SPECIAL PIPING ITEMS													
107	INSULATING JOINT	ASME B31.8	36"	--	--	--	900#	ASME 31.8 ASME VIII DIV. 1	BODY: API 5L GR.B; PUPS: API 5L GR.B, PSL2, SAWL; FILLER ISOLATOR: EPOXY RESIN COLD CURED; COATING: AMINE-CURED EPOXY	PCS	1	0	1
PIPELINE VALVES - AKABEL RECEIVER													
S/NO	SIZE	MATERIAL CLASS	VALVE TYPE	VALVE CODE	BORE TYPE	END CONNECTION	PRESSURE CLASS	DESIGN CODE	ITEM & MATERIAL DESCRIPTION	UNIT	REQUIRED QUANTITY	CONTINGENCY	TOTAL QUANTITY TO PROCURE
1	36"	ASME B31.8	BALL	—	FB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, HNDWHL OPR W/ GEARBOX	PCS	4	0	4
8	2"	ASME B31.8	BALL	—	RB	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API SPEC 6D, ASTM A105N OR ASTM A216 WCB BODY, ASTM A105N BALL WITH ENP, SPB, TRUNNION MTD BALL, LEVER OPR	PCS	4	0	4
11	3/4"	ASME B31.8	BALL	—	—	SW WITH 100mm PUP PIECE	1500#	—	SW, CL1500, API SPEC 6D, A105N BODY, SS316 TRIM, FLOATING BALL, LEVER OPR	PCS	1	0	1
14	2"	ASME B31.8	GLOBE	—	—	RTJ FLANGED WITH SMOOTH FINISH	900#	—	FLGD, RJ, CL900, API STD 623, ASME B16.10, A216 WCB BODY, TRIM 5, OS&Y, BB, SWIVEL PLUG DISC, HNDWHL OPR W/ GEARBOX	PCS	1	0	1

INDUCTION BEND LIST. PIPELINE FROM BUBOUWE MANIFOLD STATION TO ELEPA MANIFOLD STATION									
SN	PIPE ND	ANGLE (DEGRE E)	MATERIAL	WALL THICKNE SS (mm)	COATING/ THICKNESS	KP (Km)	NORTHING (m)	EASTING (m)	LOCATION
1	24"	30	API 5L GR X70	15.88	3LPE/3mm	0+000	494208.764	179065.951	BUBOUWE MANIFOLD STATION
2	24"	30	API 5L GR X70	15.88	3LPE/3mm	0+000	494209.287	179066.566	BUBOUWE MANIFOLD STATION
3	24"	75	API 5L GR X70	15.88	3LPE/3mm	0+251	494371.818	179257.006	INFIELD
4	24"	90	API 5L GR X70	15.88	3LPE/3mm	6+988	490530.889	184791.027	INFIELD
5	24"	35	API 5L GR X70	15.88	3LPE/3mm	7+067	490596.545	184836.595	INFIELD
6	24"	90	API 5L GR X70	15.88	3LPE/3mm	7+314	490842.142	184835.702	INFIELD
7	24"	30	API 5L GR X70	15.88	3LPE/3mm	7+321	490843.035	184818.54	ELEPA MANIFOLD STATION
8	24"	30	API 5L GR X70	15.88	3LPE/3mm	7+321	490843.037	184826.388	ELEPA MANIFOLD STATION

MTO for Long Lead Items (Piping/Mechanical/Pipeline)

INDUCTION BEND LIST PIPELINE FROM ELEPA MANIFOLD STATION TO AKABEL MANIFOLD STATION									
SN	PIPE ND	ANGLE (DEGRE E)	MATERIAL	WALL THICKNE SS (mm)	COATING/ THICKNESS	KP (Km)	NORTHING (m)	EASTING (m)	LOCATION
1	36"	30	API 5L GR X70	23.83	3LPE/3mm	0+000	490847.441	184828.562	ELEPA MANIFOLD STATION
2	36"	30	API 5L GR X70	23.83	3LPE/3mm	0+000	490847.406	184834.527	ELEPA MANIFOLD STATION
3	36"	90	API 5L GR X70	23.83	3LPE/3mm	0+010	490847.387	184839.899	INFIELD
4	36"	55	API 5L GR X70	23.83	3LPE/3mm	0+361	490496.969	184839.899	INFIELD
5	36"	34	API 5L GR X70	23.83	3LPE/3mm	3+581	488660.921	187485.282	INFIELD
6	36"	51	API 5L GR X70	23.83	3LPE/3mm	14+855	488421.13	198756.58	INFIELD
7	36"	34	API 5L GR X70	23.83	3LPE/3mm	19+356	484882.923	201538.782	INFIELD
8	36"	32	API 5L GR X70	23.83	3LPE/3mm	25+227	483106.542	207134.625	INFIELD
9	36"	30	API 5L GR X70	23.83	3LPE/3mm	31+376	478456.149	211155.233	AKABEL MANIFOLD STATION
10	36"	30	API 5L GR X70	23.83	3LPE/3mm	31+376	478451.284	211159.476	AKABEL MANIFOLD STATION